

Title

Clinical outcomes and implementation characteristics of adapted Pal-Cycles intervention during stepped-wedge rollout

Abstract

Background: Following successful adaptation of Pal-Cycles across seven European countries, this analysis presents clinical and implementation outcomes during stepped-wedge implementation. **Objective:** To evaluate preliminary clinical effectiveness and implementation metrics following adapted intervention deployment. **Methods:** Twelve-month outcome data were collected from 180 advanced cancer patients and their providers across all seven countries during stepped-wedge phase. Patient-reported measures included symptom burden, care satisfaction, and quality of life. Provider-level metrics assessed care coordination perception and workflow integration. Implementation fidelity was monitored through audits of care documentation and intervention protocol adherence. **Results:** Patient-reported care coordination improved by 3.2 points on an 11-point scale (95% CI 1.8-4.6). Symptom burden decreased by a mean 2.1 points on the Edmonton Symptom Assessment System ($P = 0.042$). Provider satisfaction with care planning processes increased across countries (mean 6.8/10 baseline to 8.1/10 post-implementation). Implementation fidelity averaged 81% across sites (range 74-91%). No serious intervention-related adverse events were identified. **Conclusion:** Adapted Pal-Cycles implementation demonstrated feasibility, positive preliminary clinical outcomes, and moderate-to-high fidelity across diverse European healthcare settings.